

ABHINA ANIL

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Mechanical Engineering Graduate | AI in Manufacturing | SolidWorks Certified | CGPA 8.84

PROFESSIONAL SUMMARY

B. Tech Mechanical Engineering graduate (Minor: Computer Science) at TKM College of Engineering with a CGPA of 8.84. Currently interning at Trane Technologies on an AI-driven parts management system - validating AI attribute extraction from mechanical drawings and defining critical component parameters for industrial procurement automation. Combines strong mechanical design skills (SolidWorks, ANSYS, AutoCAD) with practical AI/ML project exposure, making a unique fit for Industry 4.0, digital manufacturing, and smart engineering roles.

EDUCATION

B. Tech in Mechanical Engineering (Minor: Computer Science)

Nov 2022 – May 2026

TKM College of Engineering, Kollam | CGPA: 8.84 / 10

Relevant Coursework: Design of Machine Elements, Dynamics and Design of Machinery, Advanced Manufacturing Engineering, Management for Engineers, Product Design and Development, Sustainable Engineering, Industrial and Systems Engineering.

INTERNSHIP EXPERIENCE

Engineering Intern - AI in Parts Management

Jan 2026 – Present

Trane Technologies | India

- Working on an AI-driven parts management and procurement system. Interpreted complex mechanical engineering drawings to identify and verify attributes extracted by AI/ML models for accuracy in component classification
- Defined and prioritised critical part attributes (dimensions, tolerances, material grades, standards) to improve AI extraction performance in an industrial context
- Contributed to bridging the gap between domain engineering knowledge and AI system design in a Fortune 500 manufacturing environment

UAV/Drone Technology Intern

Jun 2025 – Jul 2025

NIT Calicut | Kerala, India

- Designed and optimised multiple drone mechanisms using SolidWorks and Fusion 360
- Conducted structural analyses in ANSYS, validating designs for robustness and performance
- Enhanced skills in 3D modelling, simulation, and engineering research through iterative design processes

Trainee and Intern - Research

Nov 2024 – Dec 2024

Antaragni '24, IIT Kanpur

- Conducted an independent research project on Materials for Radiation and Thermal Protection in Space and High-Altitude Aviation, focusing on thermal conductivity, structural resilience, and aerospace applications

Engineering Intern

Sept 2024 – Oct 2024

Cochin Port Authority | Cochin, India

- Shadowed operations in port logistics, cargo handling equipment, and vehicle maintenance
- Documented maintenance workflows and learned mechanical systems integration in large-scale maritime operations

Engineering Intern

Aug 2024 – Sept 2024

Kerala Electrical & Allied Engineering Co. Ltd. (KEL) | Kollam, India

- Gained practical exposure to industrial electrical standards and quality assurance processes
- Prepared technical documentation and assisted in quality checks for alternators, transformers, and power systems

KEY PROJECTS

AI-Driven Parts Management System - Trane Technologies Internship Project

- Developed an understanding of AI/ML attribute extraction from engineering drawings for automated procurement classification
- Identified and validated key mechanical attributes (geometry, material, tolerance, function) to improve model performance
- Architecture involved LLM/RAG-based classification pipeline; applied K-Means clustering concepts for parts categorisation

Hydrogen Production via Advanced Water Splitting - *Final Year B. Tech Project*

- Comprehensive research and development project on green hydrogen generation using advanced electrochemical and thermochemical water-splitting techniques
- Covered thermodynamic analysis, system design, electrolyser modelling, and material selection for catalysts and membranes
- Report includes chemical equations, efficiency calculations, and comparison of PEM, alkaline, and solid oxide technologies

Real-Time Weld Defect Detection using AI/ML - *Minor Project*

- Designed an AI/ML-based computer vision system to detect weld defects in real time during manufacturing
- Applied image processing and machine learning classification techniques to identify defect types with high accuracy
- Project demonstrates direct application of AI in manufacturing quality control - a key Industry 4.0 use case

ATV Roll Cage Design - *BAJA SAE*

- Designed and analysed an ATV roll cage using CAD and FEA tools (SolidWorks, ANSYS) with focus on safety, crash performance, and weight optimisation
- Applied structural mechanics principles; iterated design across multiple simulation cycles

SKILLS AND TOOLS

CAD / FEA	SolidWorks (Certified Associate), ANSYS, AutoCAD, Fusion 360
Programming	Python, C, MATLAB
AI / ML	AI attribute extraction, LLM/RAG concepts, K-Means clustering, NLP basics
Office Tools	Microsoft Excel, PowerPoint, Word
Languages	English (Fluent), Hindi (Intermediate), Malayalam (Native), Sanskrit (Elementary)

CERTIFICATIONS

- SolidWorks Certified Associate - Dassault Systèmes
- Aircraft Stability and Control - NPTEL
- Sustainable Energy Technology - NPTEL
- Junior MBA - Startup and Business Strategy - Clever Harvey (Nationally Selected Cohort)

LEADERSHIP EXPERIENCES

- **Head of Roll Cage Design** | BAJA SAE - TKM College of Engineering
- **Program Coordinator** | Society of Automotive Engineers (SAE) - TKM College of Engineering
- **Chief Financial Officer** | Innovation and Entrepreneurship Development Cell - TKM College of Engineering
- **Documentation Head** | Eureka Ideation Club, SAE - TKM College of Engineering
- **Member** | Global Shapers Community - World Economic Forum Initiative
- **Program Coordinator** | Civil Service Aspirants' Club - TKM College of Engineering
- **Class Representative** | TKM College of Engineering

COMPETITIONS AND ACHIEVEMENTS

- HULT Prize 2024 - Pitched algae-based biofuel startup concept at international entrepreneurship competition
- Smart India Hackathon 2023 - Pitched smart street light initiative for national-level government hackathon
- HESTIA Mechanical Exhibition - Built and demonstrated a functional differential model